Adaptive Methods in Clinical Research
Practical 1: Single-arm binary outcome designs

In this practical, we will obtain different types of single-arm binary outcome designs that are suitable for a planned study, and compare them.

Note: For this session we will use the `clinfun` package (on CRAN) and the `curtailment` package. The simplest way to install and load these is to first install the `librarian` package, then use its `shelf` command to install and load the required packages:

```
install.packages("librarian")
```

```
librarian::shelf(clinfun, martinlaw/curtailment)
```

You will then be able to install and load all other packages for the course in the same way (e.g., `librarian::shelf(package1, package2, package3)`). Note: this installs the latest version of `curtailment` from github, but an earlier version package is available on CRAN.

An investigator would like your advice regarding planning a trial of a new treatment for pulmonary arterial hypertension, a rare disease that has limited treatment options.

The investigator considers a response rate of 0.05 to be poor (p_0), and would like to control the type-I error-rate (α) to be at most 0.05 at this poor response rate. He thinks the response rate for the new treatment will be 0.30 (p_1), and would like to power the trial for this magnitude of response rate. The power $(1-\beta)$ must be at least 80%.

1. Use the `ph2single` command from the `clinfun` package to obtain a selection of single-stage designs for this study.

- What is the smallest sample size returned?
- What is the type-I error-rate and power of the design with the smallest sample size?

```
ph2single(pu = 0.05, pa = 0.3, ep1 = 0.05, ep2 = 1-0.8)
```

```
##    n r Type I error Type II error
## 1 14 2   0.03005364    0.16083576
## 2 15 2   0.03620024    0.12682771
## 3 16 2   0.04293785    0.09935968
## 4 18 3   0.01087322    0.16455048
## 5 19 3   0.01323601    0.13317100
```

```
# The design with the smallest sample size is in row 1, with N=14, type-I
# error-rate 0.03 and power 100*(1-0.16)=84%.
```

Now use the `ph2simon` command from the `clinfun` package to find the Simon two-stage designs for this study that would minimise the maximum sample size and minimise the expected sample size under the null hypothesis. You will also obtain any sets of admissible design parameters.

2. Given the choice, which of these designs so far (among the choice of single-stage or two-stage and the choice of design parameters) would you select for this study, and why?

```
ph2simon(pu = 0.05, pa = 0.3, ep1 = 0.05, ep2 = 1-0.8)
```

```
##
```

```
##  Simon 2-stage Phase II design
##
## Unacceptable response rate:  0.05
## Desirable response rate:  0.3
## Error rates: alpha =  0.05 ; beta =  0.2
##
##             r1 n1 r  n EN(p0) PET(p0)   qLo   qHi
## Minimax      0  7 2 14  9.112  0.6983 0.421 1.000
## Admissible   0  6 2 15  8.384  0.7351 0.129 0.421
## Optimal      0  5 2 18  7.941  0.7738 0.000 0.129
```

```
# Personal opinion.
# Single-stage design is the most simple to run.
# Minimax has the same N_max as the single-stage, but lower ESS(H0).
# Optimal has lowest ESS(H0) (7.9, vs 9.1 for minimax), but greater N_max (18 vs 14).
# Admissible has a low ESS(H0) (8.4) with only a slight increase in N_max (15 vs 14).
```

You ask the investigator for their opinion regarding the importance of minimising the maximum sample size versus minimising the expected sample size under the null hypothesis. They respond that they would like to give them equal weight when considering which set of design parameters to select.

3. With this in mind,

- What are the design parameters of the most appropriate design for this study?
- For this design, what is the probability that the trial will end early if the response rate is equal to 0.05?

```
# For each design, the values qLo and qHi denote the interval of weights for n_max
# for which the trial is optimal. Allocating equal weighting to n_max and ESS(H0)
# means assigning a weight of (omega=)0.5 to n_max.

# As such, the most appropriate design is the design that contains 0.5 in the
# [qLo, qHi] interval. In this case, this is the Minimax design, which has a
# [qLo, qHi] interval of [0.421, 1.000]. The parameters of this design are

# r1=0, n1=7, r=2, n=14.

# The probability that the trial will end early if the response rate is equal
# to 0.05 is PET(p0)=0.6983.
```

The investigator confesses that his earlier thoughts on the response rate was a guess, but he assures you that his current Phase I trial will give him a better idea of the true response rate. The results of that trial are not yet available, but he's certain that the observed response rate will be at least 0.2 and will be no greater than 0.4. He adds that the recruitment rate will be slow, around two participants per month, and the treatment is relatively expensive.

4. At *each* of these extremes for response rate,

- What Simon design would be most appropriate?
- How does each design differ from the "best" Simon two-stage design you chose above?

```
# For p1=0.2:
response.lo <- ph2simon(pu = 0.05, pa = 0.2, ep1 = 0.05, ep2 = 1-0.8)
response.lo
```

```
##
##  Simon 2-stage Phase II design
##
```

```
## Unacceptable response rate:  0.05
## Desirable response rate:  0.2
## Error rates: alpha =  0.05 ; beta =  0.2
##
##             r1 n1 r  n EN(p0) PET(p0)   qLo   qHi
## Minimax      0 13 3 27  19.81  0.5133 0.597 1.000
## Admissible   0 11 3 28  18.33  0.5688 0.414 0.597
## Optimal      0 10 3 29  17.62  0.5987 0.000 0.414
```

```r
# For p1=0.2, the most appropriate design for the weight (0.5) is the Admissible design.
# This design has a greater N_max and ESS(H0) compared to the earlier Simon design.
```

```r
# For p1=0.4:
response.hi <- ph2simon(pu = 0.05, pa = 0.4, ep1 = 0.05, ep2 = 1-0.8)
response.hi
```

```
##
##  Simon 2-stage Phase II design
##
## Unacceptable response rate:  0.05
## Desirable response rate:  0.4
## Error rates: alpha =  0.05 ; beta =  0.2
##
##         r1 n1 r n EN(p0) PET(p0)   qLo   qHi
## Minimax  0  5 1 7  5.452  0.7738 0.415 1.000
## Optimal  0  4 1 8  4.742  0.8145 0.000 0.415
```

```r
# For p1=0.4, the most appropriate design for the weight (0.5) is the minimax design.
# This design has a lower N_max and ESS(H0) compared to the earlier Simon design.
```

The investigator is still satisfied overall with using an anticipated response rate of 0.3. However, he has heard from a colleague that his trial could be made even more efficient by using a two-stage design that allows stopping for benefit.

5. Use the `find2stageDesigns` command from the `curtailment` package to find Mander and Thompson two-stage designs, which permit stopping for benefit. Compare the Mander & Thompson design with the lowest maximum sample size to your preferred Simon design from Questions 2/3. Use the same values for the desirable and undesirable response rates and type-I error-rate and power as Question 2.

```r
mander <- find2stageDesigns(n.max=,
                            p0=,
                            p1=,
                            alpha=,
                            power=,
                            benefit=)  # allow stopping for benefit (TRUE or FALSE)
```

```r
mander <- find2stageDesigns(n.max=25,
                            p0=0.05,
                            p1=0.3,
                            alpha=0.05,
                            power=0.8,
                            benefit = TRUE)
mander
```

```
## $input
##   nmin nmax   p0  p1 alpha power maxthetaF
## 1    4   25 0.05 0.3  0.05   0.8        NA
```

```
##
## $all.des
##   n1 n2  n r1 r   alpha  power EssH0   Ess e1 thetaF
## 1  6  8 14  0 2 0.04606 0.8051 7.857 8.420  1 0.4482
## 2  5 12 17  0 2 0.04669 0.8013 7.444 9.322  1 0.7472
##
## attr(,"class")
## [1] "curtailment_2stage"
```

```
# The Mander & Thompson design with the lowest maximum sample size is similar to
# the best design from Question 2/3: same max sample size (14), same ESS(H0)
# (7.9 to 1 d.p.) and same futility boundaries (0 at interim, 2 at end).
```

6. How do the two designs (Mander & Thompson from Question 5 and Simon design from Question 2/3) compare in terms of expected sample size under $p = p_1$? Which design would you choose? This is not provided in the output from the `clinfun` package, so use `find2stageDesigns` to obtain the Simon design.

```
simon <- find2stageDesigns(n.max=25,
                           p0=0.05,
                           p1=0.3,
                           alpha=0.05,
                           power=0.8,
                           benefit = FALSE) # The difference
simon
```

```
## $input
##   nmin nmax   p0  p1 alpha power maxthetaF
## 1    4   25 0.05 0.3  0.05   0.8        NA
##
## $all.des
##   n1 n2  n r1 r   alpha  power EssH0   Ess thetaF
## 1  7  7 14  0 2 0.02743 0.8101 9.112 13.42 0.3529
## 2  6  9 15  0 2 0.03005 0.8100 8.384 13.94 0.5372
## 3  5 13 18  0 2 0.03917 0.8060 7.941 15.82 0.7975
##
## attr(,"class")
## [1] "curtailment_2stage"
```

```
# The Simon design with N=14 has ESS(H1)=13.4 compared to 8.4 for the Mander
# & Thompson design.
```

7a. The data from the Phase I trial are analysed, and the estimated response rate is much smaller than anticipated: instead of 0.30, the response rate was 0.20. Use the `find2stageDesigns` command to obtain possible Simon designs and choose an appropriate design from these. (Note: the designs have a maximum sample size of less than 30).

```
simon.p20 <- find2stageDesigns(n.max=30,
                               p0=0.05,
                               p1=0.2,
                               alpha=0.05,
                               power=0.8,
                               benefit = FALSE)
simon.p20
```

```
## $input
##   nmin nmax   p0  p1 alpha power maxthetaF
```

```
## 1    4   30 0.05 0.2  0.05   0.8        NA
##
## $all.des
##   n1 n2  n r1 r   alpha  power EssH0   Ess thetaF
## 1 13 14 27  0 3 0.04159 0.8011 19.81 26.23 0.3018
## 2 11 17 28  0 3 0.04407 0.8011 18.33 26.54 0.4511
## 3 10 19 29  0 3 0.04683 0.8011 17.62 26.96 0.5449
##
## attr(,"class")
## [1] "curtailment_2stage"
```

7b. Find a stochastic curtailment design that satisfies the same requirements (in terms of type-I error-rate, power, p_0 and p_1), using the `singlearmDesign` command. Simon designs may not stop until data are available for n_1 participants. To find only stochastic curtailment designs that do not stop earlier than your chosen Simon design, use the `minstop` argument: this argument sets the earliest point at which a trial may stop. Compare the properties of both kinds of design.

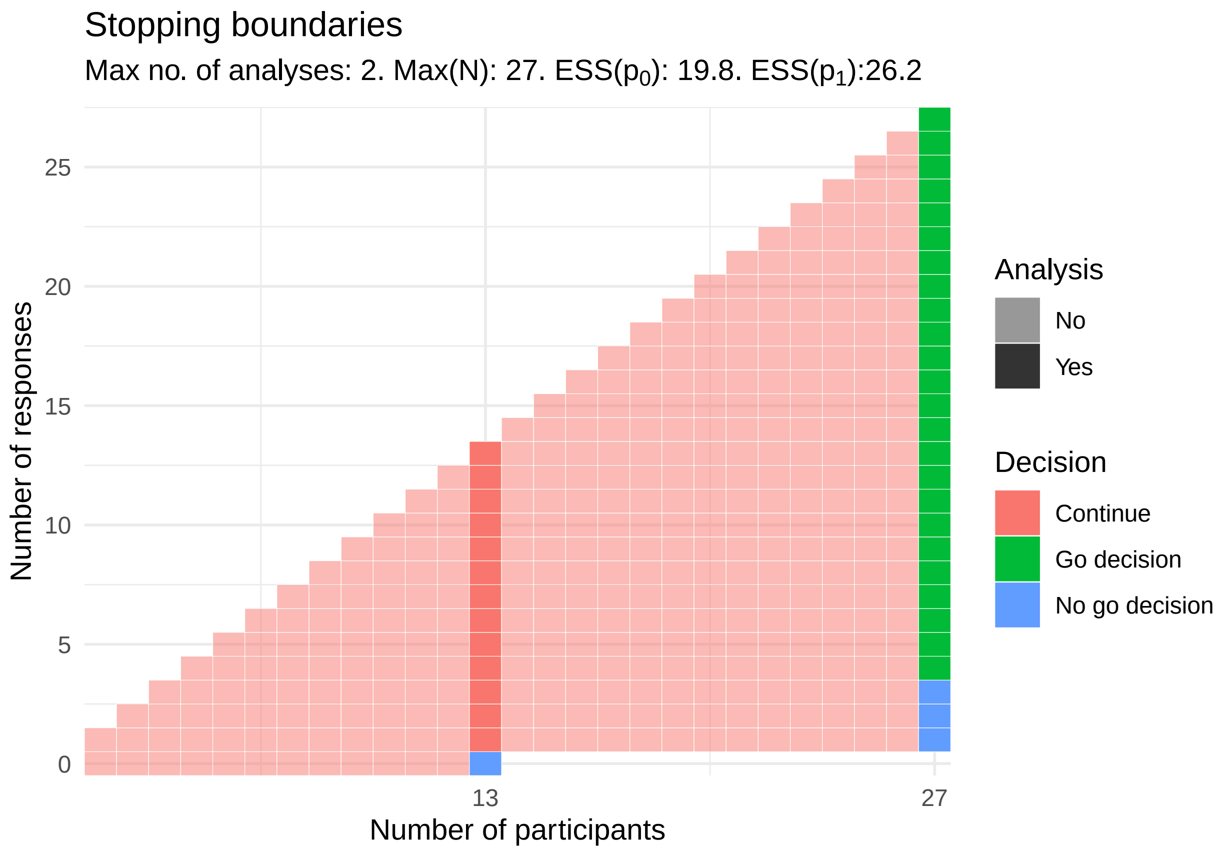
```r
sc.design <- singlearmDesign(n.max = 30,
                             p0 = 0.05,
                             p1 = 0.20,
                             power = 0.8,
                             alpha = 0.05,
                             minstop=13,
                             progressBar = FALSE)
sc.design
```

```
## $input
##   nmin nmax Cmin Cmax stagesmin stagesmax minstop   p0  p1 alpha power
## 1   13   30    1    1        NA        NA      13 0.05 0.2  0.05   0.8
##   maxthetaF minthetaE bounds fixed.r return.only.admissible max.combns
## 1       0.8       0.8   wald      NA                   TRUE      10000
##   maxthetas exact.thetaF exact.thetaE
## 1        NA           NA           NA
##
## $all.des
##        n r C   alpha  power EssH0   Ess  thetaF thetaE eff.n eff.minstop stage
## [1,] 26 3 1 0.04911 0.8025 20.15 17.50 0.04000 0.9313    26          13    26
## [2,] 27 3 1 0.04203 0.8076 19.89 18.70 0.05628 1.0000    27          13    27
## [3,] 28 3 1 0.04453 0.8141 18.38 18.49 0.08564 1.0000    28          13    28
## [4,] 29 3 1 0.04440 0.8040 16.91 18.09 0.12090 1.0000    29          13    29
##
## attr(,"class")
## [1] "curtailment_single"
```

7c. Examine the Simon designs and the stochastic curtailment designs visually, using the `plot` command. It requires two arguments: the output of a call to `find2stageDesigns` or `singlearmDesign`, and the row number of the design.

```r
plot(simon.p20, 1)
```

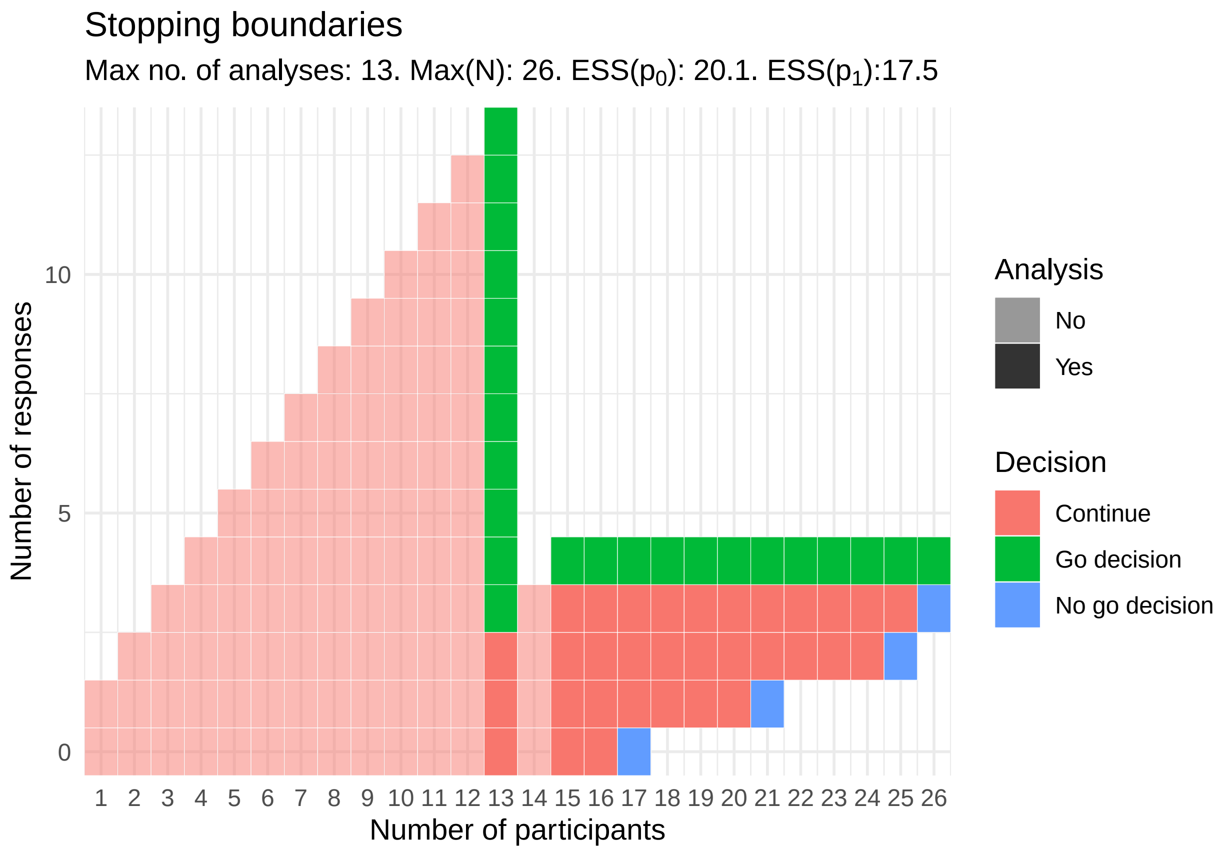
```
## $diagram
```




```
##
## $bounds.mat
##    m success fail
## 1 13     Inf    0
## 2 27       4    3
```

```
plot(sc.design, 1)
```

```
## $diagram
```




```
##
## $bounds.mat
##      m success fail
## 1    1     Inf -Inf
## 2    2     Inf -Inf
## 3    3     Inf -Inf
## 4    4     Inf -Inf
## 5    5     Inf -Inf
## 6    6     Inf -Inf
## 7    7     Inf -Inf
## 8    8     Inf -Inf
## 9    9     Inf -Inf
## 10  10     Inf -Inf
## 11  11     Inf -Inf
## 12  12     Inf -Inf
## 13  13       3 -Inf
## 14  14     Inf -Inf
## 15  15       4 -Inf
## 16  16       4 -Inf
## 17  17       4    0
## 18  18       4 -Inf
## 19  19       4 -Inf
## 20  20       4 -Inf
## 21  21       4    1
## 22  22       4 -Inf
## 23  23       4 -Inf
## 24  24       4 -Inf
```

```
## 25 25         4     2
## 26 26         4     3
```